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Study of degeneracies in pulsar noise analysis

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Introduction

Chapter 1

Gravitational waves

1.1 Gravitational waves in linearized general relativity

In general relativity, the geometry of spacetime, described by the metric $g_{\mu\nu}$, is related to the distribution of matter and energy through Einstein's field equations:

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} = 8\pi\, T_{\mu\nu}\,, \tag{1.1}$$

where $G_{\mu\nu}$ is the Einstein tensor, $R_{\mu\nu} \equiv R^{\lambda}{}_{\mu\lambda\nu}$ is the Ricci tensor, $R = g^{\mu\nu} R_{\mu\nu}$ is the Ricci scalar, and $T_{\mu\nu}$ is the stress-energy tensor [1].

In the weak-field limit, the metric can be written as a small perturbation of the flat Minkowski spacetime:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}\,, \qquad |h_{\mu\nu}| \ll 1\,, \tag{1.2}$$

where $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$ is the Minkowski metric and $h_{\mu\nu}$ is a small tensor perturbation of the metric. To first order in $h_{\mu\nu}$, indices are raised and lowered with $\eta_{\mu\nu}$.

We introduce the trace-reversed perturbation

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h\,, \qquad h = \eta^{\mu\nu} h_{\mu\nu}\,. \tag{1.3}$$

Expanding the Einstein tensor to first order in $h_{\mu\nu}$ and using the Lorenz gauge $\partial^{\mu} \bar{h}_{\mu\nu} = 0$, the field equations become

$$\Box\, \bar{h}_{\mu\nu} = -16\pi\, T_{\mu\nu}\,, \tag{1.4}$$

where $\Box = \eta^{\mu\nu} \partial_{\mu} \partial_{\nu}$ is the flat-space d'Alembert operator.

Far from the source, Eq. (1.4) becomes

$$\Box\, \bar{h}_{\mu\nu} = 0\,, \tag{1.5}$$

which is a wave equation whose solutions propagate at the speed of light.

In vacuum, the residual gauge freedom can also be used to impose the transverse-traceless (TT) gauge, defined by

$$h_{0\mu} = 0\,, \qquad h^{i}{}_{i} = 0\,, \qquad \partial^{j} h_{ij} = 0\,. \tag{1.6}$$

In the TT gauge the perturbation is traceless, so that $\bar{h}_{\mu\nu} = h_{\mu\nu}$. For a monochromatic plane wave travelling along the z-axis, the solution then takes the form

$$h^{\mathrm{TT}}_{\mu\nu}(t,z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \cos\left[\omega(t-z)\right]. \tag{1.7}$$

Here h_+ and $h_\times$ are the amplitudes of the two independent polarizations of the wave: as for electromagnetic waves, only two polarization states are allowed. Since all the t and z components are zero, the wave only acts on the plane perpendicular to its direction of propagation. The + polarization stretches distances along the x-axis while compressing them along the y-axis (and vice versa half a period later), whereas the $\times$ polarization produces the same deformation rotated by 45°.

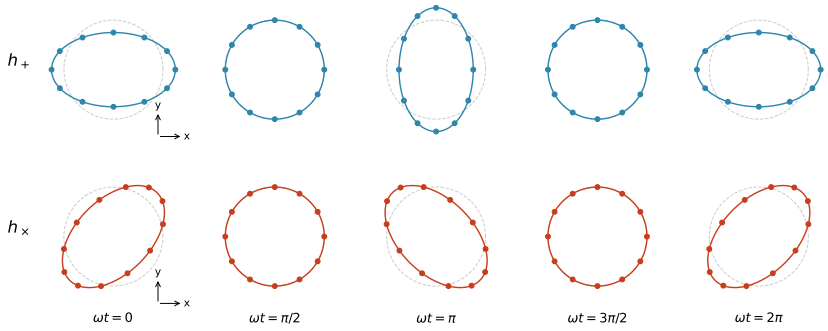


FIGURE 1.1: Deformation of a ring of test masses in the xy plane over one period, for a gravitational wave propagating along z with polarization + (top) and $\times$ (bottom).

1.2 Gravitational waves from binary systems

As in electromagnetism, the radiation field of a source can be expanded in multipoles. We consider a source far from the observer and moving slowly ($v \ll c$). The monopole and dipole terms make no contribution to the radiation, as a consequence of stress-energy conservation, which implies the conservation of the total mass-energy and of the total momentum of the system, respectively.

The leading contribution to the radiation is therefore the quadrupole term [1],

$$h_{ij}^{\mathrm{TT}} \sim \frac{2G}{c^4}\,\frac{\ddot{Q}_{ij}}{r}\,, \qquad Q_{ij} = \int \mathrm{d}^3x\,\rho\left(x_i x_j - \frac{1}{3}\,|\mathbf{x}|^2\delta_{ij}\right), \tag{1.8}$$

where Q_{ij} is the reduced (traceless) quadrupole moment of the mass density ρ of the source.

The most important sources of gravitational waves are binary systems of compact objects, whose orbital motion produces a time-varying quadrupole moment. For a binary on a circular orbit with orbital angular frequency ω_s, the mass distribution returns to the same configuration after half an orbit, so that the quadrupole moment oscillates twice per orbital period. As a consequence, gravitational waves are emitted at twice the orbital frequency,

$$\omega_{\mathrm{gw}} = 2\,\omega_s\,, \qquad \Rightarrow \qquad f_{\mathrm{gw}} = \frac{\omega_{\mathrm{gw}}}{2\pi} = 2f_{\mathrm{orb}}\,. \tag{1.9}$$

Using Kepler's law to eliminate the orbital radius, the amplitude of the emitted wave, averaged over the orientations of the binary, can be written as [2]

$$h = \sqrt{\frac{32}{5}}\,\frac{(G\mathcal{M}_c)^{5/3}\,(\pi f_{\mathrm{gw}})^{2/3}}{c^4\,r}\,, \tag{1.10}$$

where r is the distance to the source and $\mathcal{M}_c = (m_1 m_2)^{3/5}/(m_1+m_2)^{1/5}$ is the chirp mass of the binary, with m_1 and m_2 the masses of the two components. At this order, the amplitude depends on the masses only through the combination $\mathcal{M}_c$.

The total energy of the orbit can be written as a function of the gravitational-wave frequency,

$$E_{\mathrm{orb}} = -\frac{1}{2}\,\mathcal{M}_c\,(\pi G\mathcal{M}_c f_{\mathrm{gw}})^{2/3}\,, \tag{1.11}$$

while the power radiated in gravitational waves, obtained from the quadrupole formula, is

$$P = \frac{32}{5}\,\frac{c^5}{G}\left(\frac{\pi G\mathcal{M}_c f_{\mathrm{gw}}}{c^3}\right)^{10/3}\,. \tag{1.12}$$

Since the energy carried away by the waves is drawn from the orbit, $P = -\mathrm{d}E_{\mathrm{orb}}/\mathrm{d}t$, combining these two equations gives the evolution of the frequency,

$$\dot{f}_{\mathrm{gw}} = \frac{96}{5}\,\pi^{8/3}\left(\frac{G\mathcal{M}_c}{c^3}\right)^{5/3} f_{\mathrm{gw}}^{11/3}\,. \tag{1.13}$$

We note that binaries emitting at low frequencies evolve very slowly.

Since the characteristic frequency of the emission scales inversely with the total mass of the system, the binaries emitting at the lowest frequencies are supermassive black hole binaries (SMBHBs), with masses of 10^8–$10^{10}\,M_\odot$. Supermassive black holes reside at the centres of most galaxies and cannot be

the product of stellar evolution. It is widely accepted that SMBHBs form as a natural consequence of galaxy mergers: when two galaxies merge, their central black holes form a binary system which, once it reaches separations of the order of milliparsecs, emits gravitational waves in the nanohertz band [2, 3].

1.3 Stochastic gravitational wave background

A large population of compact binaries, such as SMBHBs, emitting at comparable frequencies and with comparable amplitudes cannot be resolved individually. The resulting signal is an incoherent superposition of the individual waves, which forms a stochastic gravitational wave background (GWB). A GWB is defined by the fact that the signal $h(t)$ is a random process, which cannot be described by a deterministic waveform but only statistically, through its ensemble averages. We will make the following assumptions on the GWB [1]:

- **stationary**: its statistical properties do not change in time;
- **Gaussian**: it is fully characterized by its two-point correlator;
- **isotropic**: its statistical properties are the same in every direction;
- **unpolarized**: the two polarizations are uncorrelated and have the same intensity.

Under these assumptions, the background is completely characterized by a single function of frequency, its one-sided power spectral density (PSD) $S_h(f)$, which describes how the power of the signal is distributed in frequency. From $S_h(f)$ it is convenient to define the characteristic strain [2]

$$h_c(f) = \sqrt{f\, S_h(f)}\,, \tag{1.14}$$

a dimensionless quantity that can be directly compared with the strain amplitudes of individual sources discussed in Section 1.2.

Assuming that all binaries are on circular orbits and that their evolution is driven only by the emission of gravitational waves, the energy radiated per logarithmic frequency interval follows from Eqs. (1.12) and (1.13), $f_{\rm gw}\, P/\dot{f}_{\rm gw} \propto f_{\rm gw}^{2/3}$. Summing over the whole population, the characteristic strain spectrum of the background follows the power law [2, 4]

$$h_c(f) = A_{\rm GWB} \left(\frac{f}{f_{\rm yr}}\right)^{-2/3}, \tag{1.15}$$

where $A_{\rm GWB}$ is the amplitude at the reference frequency $f_{\rm yr} = 1\,{\rm yr}^{-1}$. While the spectral index is fixed by the physics of the individual binaries, the amplitude depends on the properties of the SMBHB population, and is of the order of 10^{-15} [5–8]. In Chapter 2 we will see how this strain spectrum translates into a red noise process in the timing residuals, common to all pulsars.

Chapter 2

Pulsar Timing Array

Pulsar timing arrays (PTAs) [9–11] are currently the most sensitive method for detecting gravitational waves in the nanohertz band, where the signal is expected to be dominated by the population of supermassive black hole binaries (SMBHBs) discussed in Section 1.2.

2.1 Pulsar timing

The PTA technique relies on pulsars, which are rapidly rotating and highly magnetized neutron stars [12] that emit beams of radio radiation along their magnetic axis. Pulsars are commonly classified according to their rotational period into classical pulsars, with periods of the order of one second, and millisecond pulsars (MSPs) [13], with periods of a few milliseconds. MSPs have a much more stable rotation, which allows the times of arrival (TOAs) of their pulses to be measured with higher precision.

The measurement of the TOAs involves several steps [2, 14]. The radio emission of the pulsar propagates through the ionized interstellar medium (ISM), whose refractive index depends on frequency. As a consequence, the signal suffers a frequency-dependent delay proportional to ν^{-2}, where ν is the observing radio frequency, so that higher radio frequencies arrive earlier than lower ones. This dispersion can be corrected in two ways. In incoherent dedispersion, the observed band is divided into narrow frequency channels, and each channel is shifted in time according to the ν^{-2} law before summing them, so that the pulse is aligned across the whole band. In coherent dedispersion, the raw signal is instead convolved with the inverse transfer function of the ISM, which removes the dispersive delay exactly.

Individual pulses are usually too weak to be distinguished from the background noise. To overcome this problem, the data are divided into segments of length equal to the pulsar period and summed together, a procedure known as folding. Since the pulsed signal always falls at the same rotational phase, it adds up coherently, while the noise averages out, so that the signal-to-noise ratio grows as the square root of the number of folded pulses. The result is an average pulse profile which, unlike individual pulses, is remarkably stable over time, and which is used to determine the TOA.

The TOA is then obtained by cross-correlating the folded profile with a high signal-to-noise template of the same pulsar. The resulting phase shift, converted into time and added to the reference time of the observation, gives the arrival time of the pulse together with its uncertainty.

Since the TOAs are measured at the observatory, which is not an inertial reference frame, they are then referred to the Solar System barycenter (SSB) through a series of clock and propagation corrections. The barycentric TOAs are finally compared with the predictions of a timing model, which describes the rotation of the pulsar together with its astrometric and, for binary systems, orbital parameters. In this work we use the timing solution of the EPTA DR2low dataset [15], which combines the EPTA second data release [16] with low-frequency observations from LOFAR. It was obtained by refitting the EPTA timing model to the combined data with software based on TEMPO2 [17, 18]. The difference between the observed and the predicted TOAs defines the timing residuals,

$$\delta t_i = t^{\mathrm{obs}}_{\mathrm{TOA},i} - t^{\mathrm{model}}_{\mathrm{TOA},i}. \tag{2.1}$$

Any effect not included in the timing model, such as noise processes or gravitational waves, leaves a signature in the residuals. Since the TOAs are typically collected every few weeks over a time span T of one or two decades, PTAs are sensitive to signals with frequencies between about $1/T$, which is 1.6 nHz for $T = 20\,\mathrm{yr}$, and the Nyquist frequency $1/(2\Delta t)$ set by the observing cadence Δt, which is about $4 \times 10^{-7}\,\mathrm{Hz}$ for observations every two weeks [19]. As an example, Figure 2.1 shows the timing residuals of PSR J1640+2224, the pulsar analyzed in this work, in the DR2low dataset. The residuals at low radio frequencies have a much larger scatter, partly because of the larger TOA uncertainties and partly because the effects of the ISM are stronger at low frequencies, as discussed in Chapter 3.

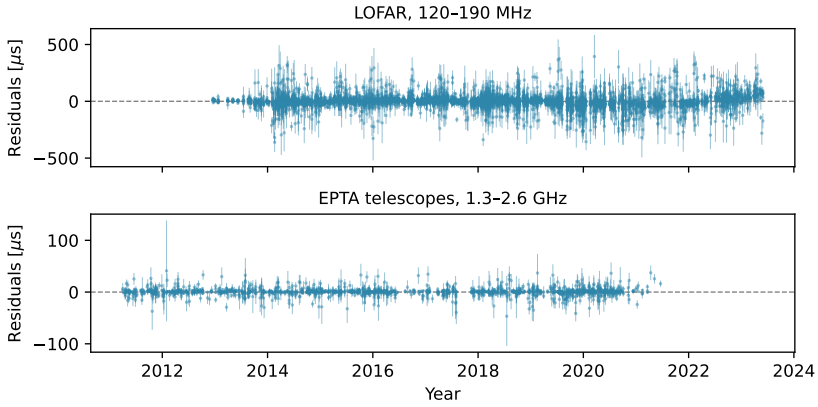


FIGURE 2.1: Timing residuals of PSR J1640+2224 in the EPTA DR2low dataset [15], for the LOFAR observations at 120–190 MHz (top) and for the EPTA telescopes at 1.3–2.6 GHz (bottom). Note the different vertical scales.

2.2 Gravitational waves in pulsar timing

A gravitational wave passing between the Earth and a pulsar perturbs the spacetime metric along the path of the radio pulses, slightly changing their travel time. As a result, the rate at which the pulses are received is shifted with respect to the intrinsic rotational frequency of the pulsar. This effect is described by the redshift [2, 10, 20]

$$z(t) \equiv \frac{\nu_0 - \nu(t)}{\nu_0}, \tag{2.2}$$

where ν_0 is the unperturbed pulse frequency and $\nu(t)$ the observed one. Since the TOAs accumulate this shift over time, the contribution of the gravitational wave to the timing residuals is given by its time integral,

$$R(t) = \int_0^t z(t')\, dt'. \tag{2.3}$$

For a stochastic background, $R(t)$ is itself a random process and cannot be predicted as a function of time. It is instead characterized by its power spectral density, which follows from the strain spectral density $S_h(f)$ introduced in Section 1.3. Two factors relate the two quantities. Averaging the response of the Earth–pulsar system over all propagation directions of the background gives a factor $1/3$, while the time integral in Eq. (2.3) introduces a factor $1/(2\pi f)^2$. Using Eq. (1.14), the PSD of the timing residuals induced by the GWB is [2]

$$S(f) = \frac{h_c^2(f)}{12\pi^2 f^3}. \tag{2.4}$$

Inserting the power-law spectrum of Eq. (1.15), expected for a population of circular SMBHBs whose evolution is driven by GW emission, we obtain

$$S(f) = \frac{A_{\rm GWB}^2}{12\pi^2} \left(\frac{f}{f_{\rm yr}}\right)^{-\gamma_{\rm GWB}} f_{\rm yr}^{-3}, \qquad \gamma_{\rm GWB} = \frac{13}{3}, \tag{2.5}$$

since $h_c^2/f^3 \propto f^{-4/3}\, f^{-3} = f^{-13/3}$. This PSD is a power law with the same functional form used to describe the intrinsic red noise of a pulsar. As a consequence, in the timing residuals of a single pulsar the GWB is indistinguishable from intrinsic red noise. The signal can only be identified by exploiting the fact that it is common to all the pulsars of the array and induces a specific correlation between their residuals.

The correlation between the timing residuals of two pulsars i and j is described by the Hellings–Downs curve [21]. Under the assumptions introduced in Section 1.3, this correlation does not depend on the positions of the pulsars on the sky, but only on their angular separation ζ_{ij},

$$\Gamma_{ij} = \frac{3}{2}\, x_{ij} \ln x_{ij} - \frac{x_{ij}}{4} + \frac{1}{2} + \frac{1}{2}\,\delta_{ij}, \qquad x_{ij} = \frac{1 - \cos\zeta_{ij}}{2}, \tag{2.6}$$

where δ_{ij} is the Kronecker delta. The cross-PSD of the residuals of pulsars i and j is then $S_{ij}(f) = \Gamma_{ij}\, S(f)$, with $S(f)$ given by Eq. (2.4), so that for a single pulsar ($\Gamma_{ii} = 1$) Eq. (2.4) is recovered.

The Hellings–Downs curve is shown in Figure 2.2. The correlation tends to $1/2$ for pulsars that are close on the sky, reaches a minimum of about -0.15 at $\zeta \simeq 82°$, and rises again to $1/4$ for pulsars in opposite directions. This angular pattern reflects the quadrupolar nature of gravitational waves and is the distinctive signature of a GWB.

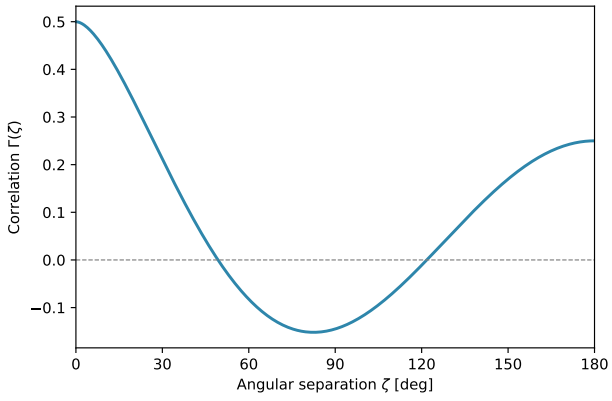


FIGURE 2.2: The Hellings–Downs curve of Eq. (2.6), giving the expected correlation between the timing residuals of two distinct pulsars as a function of their angular separation ζ.

In 2023 four PTA collaborations, EPTA with InPTA, NANOGrav, PPTA

and CPTA, reported evidence for a common red process with Hellings–Downs correlations [5–8]. Since the signal is weak and can be mimicked by the red noise of each pulsar, an accurate description of the pulsar noise is essential. The noise sources and their modeling are described in Chapter 3.

Chapter 3

The pulsar noise model

The TOAs can be described as the sum of a deterministic contribution, given by the timing model introduced in Section 2.1, and a stochastic contribution due to several noise processes [19, 22]. In this chapter we describe these noise processes and the way they are modeled. Although they have different physical origins, they are all modeled as Gaussian processes.

A first broad distinction is between white noise and red noise. White noise has a flat power spectral density, that is, the same power at all Fourier frequencies f of the timing residuals, so it produces no slow trends in the residuals. Red noise instead has more power at low frequencies, and it produces slow, time-correlated variations of the residuals [2]. When two noise processes have a similar effect on the timing residuals, the data cannot fully separate them, and part of the power of one process can be attributed to the other. Their parameters are then degenerate [19, 23].

3.1 Gaussian processes with power-law spectra

All the red noise processes considered in this work are described in the same way. A red noise process in the timing residuals is written as a Fourier series truncated to N frequencies [2],

$$\delta t_{\mathrm{red}}(t_i) = \sum_{k=1}^{N} \Big[a_k \sin(2\pi f_k t_i) + b_k \cos(2\pi f_k t_i) \Big], \qquad f_k = \frac{k}{T}, \tag{3.1}$$

where T is the time span of the data. The lowest frequency is therefore $1/T$, while the number of components N sets the highest one, N/T. Only low frequencies are needed, since red noise has most of its power there.

The coefficients a_k and b_k are independent Gaussian random variables with zero mean, whose variance is fixed by the PSD of the process,

$$\langle a_k^2 \rangle = \langle b_k^2 \rangle = \frac{S(f_k)}{T}. \tag{3.2}$$

For the PSD we use a power law with amplitude A and spectral index γ,

$$S(f) = \frac{A^2}{12\pi^2} \left(\frac{f}{f_{\rm yr}}\right)^{-\gamma} f_{\rm yr}^{-3}, \tag{3.3}$$

which for $\gamma = \gamma_{\rm GWB} = 13/3$ gives back the PSD of the GWB in Eq. (2.5). Each red noise process is therefore described by only two parameters, A and γ.

3.2 White noise

As mentioned above, white noise has a flat power spectral density. In pulsar timing analysis there are three main sources of white noise [2], each described by a parameter defined separately for every observing system, that is, every combination of receiver and backend.

- **Radiometer noise (EFAC).** The TOA uncertainty $\sigma_{{\rm TOA},i}$ obtained from the template fit may not include all sources of error, so it is multiplied by a correction factor, the EFAC.
- **Additional white noise (EQUAD).** White noise not related to the radiometer, for example due to instrumental effects, is added in quadrature to the TOA uncertainty through the EQUAD.
- **Pulse jitter (ECORR).** Since each TOA is obtained by folding a finite number of pulses, the folded profile fluctuates from one observation to the next. This noise is fully correlated between TOAs of the same observing epoch and uncorrelated between different epochs, and it is described by the ECORR.

Without ECORR, the variance of the i-th TOA is then

$$\sigma_i^2 = {\rm EFAC}^2\, \sigma_{{\rm TOA},i}^2 + {\rm EQUAD}^2. \tag{3.4}$$

In this work the white noise parameters are fixed to the values provided with the DR2low dataset, and ECORR is included only for the LOFAR systems [15].

3.3 Intrinsic red noise

The intrinsic red noise, also called spin noise, results from irregularities in the rotation of the pulsar [19]. Although MSPs have a very stable rotation, they still show small fluctuations in their rotational phase and period [2]. These fluctuations produce a red noise process that is achromatic, that is, it does not depend on the observing radio frequency ν. Its power spectral density is

modeled as a power law,

$$S_{\mathrm{RN}}(f) = \frac{A_{\mathrm{RN}}^2}{12\pi^2}\left(\frac{f}{f_{\mathrm{yr}}}\right)^{-\gamma_{\mathrm{RN}}} f_{\mathrm{yr}}^{-3}, \tag{3.5}$$

where A_{RN} is the amplitude at the reference frequency f_{yr} and γ_{RN} is the spectral index. This PSD has the same functional form as the one of the GWB in Eq. (2.5). As discussed in Section 2.2, in a single pulsar the intrinsic red noise therefore cannot be distinguished from the GWB.

The inferred parameters of the intrinsic red noise also depend on how the other noise processes are modeled. For example, NANOGrav and PPTA, which describe the DM variations with different methods, find slightly different spin noise parameters for the pulsars observed by both arrays [19].

3.4 Chromatic red noise

The radio pulses reach the Earth after travelling through the ionized interstellar medium, which delays them by an amount that depends on the observing radio frequency ν, with lower frequencies delayed more. Since the pulsar, the Earth and the ISM move with respect to each other, the electron content along the line of sight changes with time, and so do these delays [19]. The result is a red noise process in the timing residuals whose amplitude depends on ν, called chromatic red noise.

Chromatic red noise is modeled with the same Fourier basis used for the achromatic red noise, scaled by a power of the radio frequency [15, 23],

$$\delta t_{\mathrm{chrom}}(t_i, \nu_i) = \left(\frac{\nu_{\mathrm{ref}}}{\nu_i}\right)^{\alpha} \sum_{k=1}^{N} \Big[a_k \sin(2\pi f_k t_i) + b_k \cos(2\pi f_k t_i)\Big], \tag{3.6}$$

where ν_i is the radio frequency of the i-th TOA, $\nu_{\mathrm{ref}} = 1400\,\mathrm{MHz}$ is a reference frequency and α is the chromatic index. The case $\alpha = 0$ gives back the achromatic red noise, while $\alpha = 2$ and $\alpha = 4$ describe the dispersion measure variations and the scattering variations discussed below. Because of the factor $\nu_i^{-\alpha}$, these processes are much stronger at low radio frequencies.

3.4.1 Dispersion measure variations

As mentioned in Section 2.1, a radio wave travelling through the ionized ISM is delayed by an amount that depends on its frequency. For a cold plasma the dispersive delay is

$$\Delta t_{\mathrm{DM}} = \mathcal{D}\,\frac{\mathrm{DM}}{\nu^2}, \qquad \mathrm{DM} = \int_0^L n_e \,\mathrm{d}l, \tag{3.7}$$

where the dispersion measure DM is the integral of the electron density n_e along the line of sight, usually expressed in $\mathrm{pc\,cm^{-3}}$, and $\mathcal{D} = 4.148808 \times 10^3\,\mathrm{s\,MHz^2\,cm^3\,pc^{-1}}$ is the dispersion constant [19].

The mean value of the DM and its linear and quadratic trends in time are included in the timing model [2]. Because of the relative motion of the pulsar, the Earth and the ISM, however, the line of sight crosses different electron densities over time, and the remaining DM variations are stochastic. They are modeled as a chromatic red noise process, Eq. (3.6) with $\alpha = 2$, with a power-law PSD of the same form as the intrinsic red noise,

$$S_{\mathrm{DM}}(f) = \frac{A_{\mathrm{DM}}^2}{12\pi^2} \left(\frac{f}{f_{\mathrm{yr}}}\right)^{-\gamma_{\mathrm{DM}}} f_{\mathrm{yr}}^{-3}, \tag{3.8}$$

where A_{DM} is the amplitude of the delay at the reference frequency ν_{ref}.

3.4.2 Scattering variations

Inhomogeneities in the ISM also scatter the radio waves, so that part of the signal reaches the observer along longer paths. This multipath propagation broadens the pulse with an exponential tail, whose characteristic timescale τ_{sc} depends strongly on the radio frequency [19],

$$\tau_{\mathrm{sc}} \propto \nu^{-\alpha}. \tag{3.9}$$

Since the line of sight changes with time, the scattering also varies and introduces an additional chromatic red noise in the TOAs. It is modeled with Eq. (3.6) and a power-law PSD of the same form as Eq. (3.8), with amplitude A_{SV} and spectral index γ_{SV}.

The value of α depends on the properties of the scattering medium. A thin screen with inhomogeneities of the same size gives $\alpha = 4$, which is the value adopted for the DR2low dataset, while an extended medium with Kolmogorov turbulence gives $\alpha = 4.4$ [15]. In this work we use $\alpha = 4$ as the reference model, and we study the effect of $\alpha = 4.4$ and of a free chromatic index in Chapter 6.

3.4.3 Other chromatic noise processes

DM variations and scattering are not the only chromatic effects in pulsar timing. The solar wind adds a further contribution to the DM, which is largest when the line of sight passes close to the Sun and therefore varies along the year. Other effects include sudden chromatic events, such as exponential dips in the residuals, and additional delays caused by refraction in the ISM [15, 19]. For PSR J1640+2224, which lies far from the ecliptic plane, the solar wind is described only by a deterministic term in the timing model [15], and none

of these processes is included in the noise model. The only additional noise process considered in this work is the band noise, discussed in Section 3.5.

3.5 Band noise

Band noise is a red noise process that is present only in the TOAs observed within a given radio frequency band [22]. In this work the band considered is the LOFAR band, 120–190 MHz, which contains the TOAs recorded by the seven LOFAR observing systems of the DR2low dataset.

The physical origin of band noise is not known with certainty. It may come from effects of the ISM that are not coherent between different bands, for example because photons at different frequencies cross different scattering volumes, or that scale with frequency more steeply than ν^{-2}, and therefore appear as an excess of noise at low frequencies. Alternatively, it may be caused by frequency-dependent calibration errors, which can remain correlated over hundreds of days, or by radio frequency interference (RFI) present in the same band at different sites. RFI, however, is expected to be mostly uncorrelated between observations, and in that case it is absorbed by the white noise parameters. In the first IPTA data release, band noise was detected in four pulsars, with amplitudes $\log_{10} A_{\mathrm{BN}}$ between about -14 and -12 and spectral indices γ_{BN} between about 0.6 and 3.3 [22].

Band noise must be distinguished from system noise, a red noise process that affects only the TOAs of a single observing system, that is, a single combination of telescope, receiver and backend, and that can arise from instrumental effects or calibration errors. If a band contains only one system, the two models act on exactly the same TOAs and are completely covariant, so that the data cannot distinguish between them. When several systems observe in the same band, as for the seven LOFAR systems used here, a noise process common to all of them can instead be attributed to the band. For this reason, low-frequency observations such as those of LOFAR, together with wide-band receivers, are particularly useful to identify band-dependent effects [22].

The band noise is modeled as a power law of the same form as Eq. (3.5), with amplitude A_{BN} and spectral index γ_{BN}. The Fourier basis of Section 3.1 is applied only to the TOAs of the LOFAR band and is set to zero for all the other TOAs. Unlike the chromatic noise of Eq. (3.6), no factor $(\nu_{\mathrm{ref}}/\nu_i)^{\alpha}$ is included, since the dependence on the radio frequency enters only through the selection of the TOAs. We use $N = 50$ Fourier components. This number was fixed a priori and was not obtained from a model selection.

3.6 Noise model for PSR J1640+2224

In summary, the reference noise model used in this work for PSR J1640+2224 contains the white noise, the intrinsic red noise, the DM variations and the

scattering variations with $\alpha = 4$. The white noise parameters are fixed, with an EFAC and an EQUAD for each of the 13 observing systems and an ECORR for each of the seven LOFAR systems, so that the free parameters are the six amplitudes and spectral indices of the three red noise processes. In Chapter 6 we also consider three variants of this model, one with $\alpha = 4.4$, one in which α is a free parameter, and one that includes the band noise of Section 3.5. The number of Fourier components and the priors of each process are given in Section 4.1.

Chapter 4

Bayesian inference and analysis tools

The noise model described in Chapter 3 depends on a small number of parameters, the amplitudes and spectral indices of the noise processes. In this chapter we describe how these parameters are estimated from the timing residuals. We first introduce Bayesian inference, then the likelihood used in pulsar timing analysis, and finally the method used to sample the posterior distribution.

4.1 Bayesian inference

In Bayesian inference, the result of a measurement is the posterior distribution $p(\boldsymbol{\theta}|\delta\boldsymbol{t})$, that is, the probability density of the model parameters $\boldsymbol{\theta}$ given the data, which in our case are the timing residuals $\delta\boldsymbol{t}$. The posterior is given by Bayes' theorem,

$$p(\boldsymbol{\theta}|\delta\boldsymbol{t}) = \frac{\mathcal{L}(\delta\boldsymbol{t}|\boldsymbol{\theta})\,\pi(\boldsymbol{\theta})}{\mathcal{Z}}, \tag{4.1}$$

where $\mathcal{L}(\delta\boldsymbol{t}|\boldsymbol{\theta})$ is the likelihood, the probability density of the data given the parameters, and $\pi(\boldsymbol{\theta})$ is the prior, which describes what we know about the parameters before the measurement. The denominator is the evidence,

$$\mathcal{Z} = \int \mathcal{L}(\delta\boldsymbol{t}|\boldsymbol{\theta})\,\pi(\boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta}, \tag{4.2}$$

which does not depend on $\boldsymbol{\theta}$. For parameter estimation it is only a normalization constant, while it becomes important when different models are compared, as discussed in Section 4.4.

In the reference model of this work the free parameters are the amplitudes and spectral indices of the intrinsic red noise, of the DM variations and of the scattering variations, modeled with 54, 40 and 40 Fourier components respectively. For all the amplitudes we use a prior uniform in $\log_{10} A$ between -20 and -11, and for all the spectral indices a uniform prior between 0 and 7. A prior uniform in the logarithm is the usual choice for a quantity whose order of magnitude is not known. The same priors are used for the band noise,

and when the chromatic index α is left free its prior is uniform between 0 and 10.

The posterior is a function of all the parameters. To study a single parameter θ_i we integrate the posterior over all the others, obtaining the marginalized posterior

$$p(\theta_i|\delta\boldsymbol{t}) = \int p(\boldsymbol{\theta}|\delta\boldsymbol{t}) \prod_{k\neq i} \mathrm{d}\theta_k. \tag{4.3}$$

In this way the uncertainty on the other parameters is included in the result for θ_i. In particular, if two parameters are covariant, marginalizing over one of them broadens the posterior of the other.

4.2 The pulsar timing likelihood

To write the likelihood we need a model for the timing residuals. We write them as the sum of three contributions,

$$\delta\boldsymbol{t} = \boldsymbol{M}\boldsymbol{\epsilon} + \boldsymbol{F}\boldsymbol{a} + \boldsymbol{n}. \tag{4.4}$$

The first term accounts for the timing model. The timing model parameters are obtained from a fit performed before the noise analysis, and this fit may have absorbed part of the noise. Since the corrections are small, the timing model is linearized around the best-fit values. The design matrix $\boldsymbol{M}$ contains the derivatives of the TOAs with respect to the timing model parameters, and $\boldsymbol{\epsilon}$ is the vector of the small offsets of these parameters from their best-fit values. The second term contains the red noise processes of Chapter 3. The columns of $\boldsymbol{F}$ are the sines and cosines of Eq. (3.1), multiplied by the chromatic factor of Eq. (3.6) for the chromatic processes, and $\boldsymbol{a}$ is the vector of the Fourier coefficients a_k and b_k of all the processes. The last term, $\boldsymbol{n}$, is the white noise. It is Gaussian with covariance matrix $\boldsymbol{C}_{\mathrm{WN}}$, whose diagonal elements are the variances σ_i^2 of Eq. (3.4), while ECORR adds a correlation between the TOAs of the same LOFAR system observed at the same epoch.

For fixed $\boldsymbol{\epsilon}$ and $\boldsymbol{a}$, the quantity $\delta\boldsymbol{t} - \boldsymbol{M}\boldsymbol{\epsilon} - \boldsymbol{F}\boldsymbol{a}$ is only white noise, so the likelihood is a Gaussian. However, we are not interested in the values of $\boldsymbol{\epsilon}$ and $\boldsymbol{a}$, which describe one particular realization of the noise, but in the parameters $\boldsymbol{\theta}$ of the power spectral densities. The coefficients $\boldsymbol{a}$ have a Gaussian prior with the variances of Eq. (3.2), which depend on $\boldsymbol{\theta}$ through the PSD, while for $\boldsymbol{\epsilon}$ we use a uniform prior. With these choices the integral over $\boldsymbol{\epsilon}$ and $\boldsymbol{a}$ can be computed analytically, and the result is again a Gaussian,

$$\mathcal{L}(\delta\boldsymbol{t}|\boldsymbol{\theta}) = \frac{\exp\left(-\frac{1}{2}\,\delta\boldsymbol{t}^T\boldsymbol{C}^{-1}\delta\boldsymbol{t}\right)}{\sqrt{\det(2\pi\boldsymbol{C})}}, \tag{4.5}$$

with covariance matrix

$$\boldsymbol{C}(\boldsymbol{\theta}) = \boldsymbol{C}_{\mathrm{WN}} + \boldsymbol{F}\,\boldsymbol{\Phi}(\boldsymbol{\theta})\,\boldsymbol{F}^T, \tag{4.6}$$

where $\boldsymbol{\Phi}$ is the diagonal matrix of the variances of Eq. (3.2). The marginalization over $\boldsymbol{\epsilon}$ adds to $\boldsymbol{C}$ a term of the same form for the timing model, with $\boldsymbol{M}$ in place of $\boldsymbol{F}$ and a very large variance, which corresponds to the uniform prior.

This marginalized likelihood depends only on $\boldsymbol{\theta}$. Since the white noise parameters are fixed, in the reference model it is a function of six parameters.

4.3 Sampling the posterior

The posterior of Eq. (4.1) cannot be computed analytically, and evaluating it on a grid becomes too expensive as the number of parameters grows. We therefore use a Markov chain Monte Carlo (MCMC) method, which produces a chain of samples of $\boldsymbol{\theta}$ distributed according to the posterior. The marginalized posterior of a parameter is then simply the histogram of its samples.

The chain is built with the Metropolis–Hastings algorithm. It starts from a point drawn from the prior. At each iteration a new point is proposed, and the posterior at the proposed point is compared with the posterior at the current one. If it is larger the move is accepted, otherwise it is accepted with probability equal to the ratio of the two values. If the move is rejected, the chain stays at the current point. Since only ratios of the posterior are needed, the evidence does not have to be computed. We use the PTMCMCSampler package, which combines several proposal distributions that are adapted during the run using the previous samples of the chain.

Each chain consists of 10^6 iterations, of which one sample every 10 is saved. The first part of the chain depends on the starting point and does not represent the posterior, so we discard the first 25% of the samples as burn-in.

Consecutive samples of the chain are correlated, so the number of independent samples is smaller than the length of the chain. The correlation is measured by the integrated autocorrelation time τ_{ac}, expressed in units of saved samples, which we estimate for each parameter. The effective number of samples is then

$$N_{\mathrm{eff}} = \frac{N_{\mathrm{s}}}{\max_i \tau_{\mathrm{ac},i}}, \tag{4.7}$$

where N_{s} is the number of samples left after the burn-in and the maximum is taken over the parameters. We use N_{eff} to check that the chain is long enough.

The results are shown as corner plots. The panels on the diagonal show the marginalized posterior of each parameter, and the other panels show the two-dimensional marginalized posterior of each pair of parameters, with contours enclosing 68%, 95% and 99.7% of the probability. The two-dimensional panels

show the covariance between the parameters, and are therefore the main tool to identify degeneracies.

4.4 Model comparison

To compare two models we use the evidence of Eq. (4.2). The ratio of the evidences of two models is the Bayes factor,

$$\mathcal{B}_{12} = \frac{\mathcal{Z}_1}{\mathcal{Z}_2}, \tag{4.8}$$

which tells how much the data favor model 1 over model 2, if the two models are considered equally probable before the measurement. Since the evidence is the likelihood averaged over the prior, a model with additional parameters is favored only if it fits the data better by an amount that compensates for its larger prior volume.

The evidence is an integral over the whole parameter space, and it is not provided by the MCMC sampler. When one model is contained in the other, however, the Bayes factor can be obtained from the posterior alone. This is the case of the band noise. The model with band noise reduces to the model without it when the amplitude A_{BN} is negligible, which in practice corresponds to the lower bound of its prior, $\log_{10} A_{\mathrm{BN}} = -20$. The Bayes factor of the model without band noise over the model with band noise is then given by the Savage–Dickey density ratio,

$$\mathcal{B} = \frac{p(\log_{10} A_{\mathrm{BN}} = -20 \,|\, \delta\boldsymbol{t})}{\pi(\log_{10} A_{\mathrm{BN}} = -20)}, \tag{4.9}$$

where the numerator is the marginalized posterior and the denominator is the prior, both evaluated in the model with band noise. If the data require the band noise, the posterior at the lower bound is smaller than the prior and $\mathcal{B} < 1$. If instead $\mathcal{B} \simeq 1$, the data cannot distinguish between the two models. We estimate the numerator from the fraction of samples in the lowest unit interval of the prior, $-20 < \log_{10} A_{\mathrm{BN}} < -19$.

4.5 Enterprise and Discovery

The likelihood of Eq. (4.5) is implemented in Enterprise, the software commonly used in PTA analyses, which was also used for the noise analysis of the DR2low dataset. It is written in Python and relies on the NumPy and SciPy libraries for its linear algebra operations. Discovery is a more recent package that implements the same likelihood with the JAX library.

JAX is a Python library for numerical computing. A function written with JAX can be compiled just in time, that is, it is translated into optimized code the first time it is called, so that the following calls are much faster. JAX

functions can also be differentiated automatically and executed on graphics processing units (GPUs). In this work we use only the first feature. The likelihood is compiled and evaluated on CPUs in double precision, and its gradient is not needed by the Metropolis–Hastings algorithm.

In Chapter 5 we compare the posteriors obtained with the two codes for the same model and the same priors. Since both analyses use PTMCMCSampler, the comparison is a test of the implementation of the likelihood. The autocorrelation times are computed with the emcee package and the corner plots are made with the corner package.

Chapter 5

Validation of Discovery on PSR J1640+2224

Chapter 6

Study of degeneracies

Conclusions

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